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Activity - 4

Hashing

Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989}

hash function $h(x) = x \pmod{10}$;

(a) Compute hash values.

key	$h(x) = x \bmod 10$
4371	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

(a) separate chaining Hash Table:

Index	chain
0	—
1	4371
2	—
3	1323 → 6173
4	4344
5	—
6	—
7	—
8	—
9	4139 → 9679 → 1989

(b) Open Addressing using Linear Problem:

-> Insertions:-

* 4371 → 1
* 1323 → 3
* 6173 → 3 → 4
* 4344 → 4 → 5
* 4199 → 9
* 9679 → 9 → 10
* 1989 → 9 → 0 → 1 → 2

Final Table:

Index	Value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	–
7	–
8	–
9	4199

(c) Open Addre

-> Formula:

h:(

Insertions:

4371 →

1323 →

6173 →

4344 →

4199 →

9679 →

1989 →

Final Table

Index
0
1
2
3
4
5
6
7
8
9

(c) Open Addressing using

→ Formula:

$$h_i(x) = (h(x) + i^2) \bmod 10$$

Insertions:

4371 → 1

1323 → 3

6173 → 3 → 4

4344 → 4 → 5

4199 → 9

9679 → 9 → 0

1989 → 9 → 0 → 3 → 8

Find Table:

Index	Value:
0	9679
1	4371
2	–
3	1323
4	6173
5	4344
6	–
7	–
8	1989
9	4199

(d) Open Addressing using Double Hashing

Compute second Hash

Key	$x \bmod 7$	$h_2(x)$
4371	3	4
1323	0	7
6173	6	1
4199	6	1
4344	4	3
9679	5	2
1989	1	6

Insertions:

* 4371 → 1
* 1323 → 3
* 6173 → 3 → (3+1) = 4
* 4344 → 4 → (4+3) = 7
* 4199 → 9
* 9679 → 9 → (9+2) = 11
* 1986 → 9 → (9 + 6) = 15

Final table

Index	Value
0	–
1	4371
2	–
3	1326
4	6173
5	9673
6	–
7	4344
8	–
9	4199